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An abbreviated Caprini model for VTE risk assessment in trauma

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Abstract

The Caprini risk assessment model (RAM) is widely used to assess risk of venous thromboembolism (VTE). However, it is cumbersome with 31 variables and poses challenges with inter-rater reliability. This study aimed to determine if an abbreviated model could perform similarly in VTE risk assessment. We performed a retrospective review of trauma patients ≥ 18 years old and admitted for over 24 h at a Level I trauma center from January 1, 2018, to December 31, 2018. Demographic and clinical data were analyzed to generate Caprini scores. Using a p-value cutoff of < 0.05, the individual components of the original Caprini RAM most highly associated with VTE were identified and used to calculate an abbreviated Caprini score. Logistic regression assessed odds of inpatient VTE with the original or abbreviated Caprini RAMs. Receiver operating characteristic curves and c-statistics were generated to assess discriminatory ability. The study sample included 1279 patients. Ten risk factors were included in the abbreviated model (recent major surgery, length of surgery > 2 h, transfusion, restricted mobility > 72 h, central venous catheter, current major surgery, age, history of VTE, hip or leg fracture, and serious trauma). Compared to the original, the abbreviated model had a similar odds ratio (1.17 vs 1.07, both p-values < 0.001), c-statistic (0.747 vs 0.753), sensitivity (0.73 vs 0.76) and specificity (0.62 vs 0.61). An abbreviated Caprini RAM performs similarly to the original, may streamline workflow and allow for automation in electronic health records, potentially enhancing its use in resource limited settings.

Keywords Venous thromboembolism · Risk assessment · Caprini · Trauma · Thromboprophylaxis

Abbreviations

AIS	Abbreviated Injury Scale
BMI	Body mass index
CI	Confidence interval
DVT	Deep vein thrombosis
ISS	Injury Severity Score
IPC	Intermittent pneumatic compression
IVC	Inferior vena cava
PE	Pulmonary embolus
RAM	Risk assessment model
SD	Standard deviation
SICU	Surgical intensive care unit
VTE	Venous thromboembolism

Highlights

- The Caprini risk assessment model for stratification of surgical patients at risk for VTE is cumbersome, time-consuming to complete, and includes subjective variables.
- We have created an abbreviated model using the ten components from the original model that are most highly associated with VTE, also removing all subjective variables.
- In a population of trauma patients from a single institution, the abbreviated Caprini model performed similarly to the original model.
- An abbreviated model may streamline workflow and reduce need for costly laboratory testing, potentially enhancing use in resource-limited settings.

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Introduction

Venous thromboembolism (VTE) is the third most common cause of in-hospital death for trauma patients [1]. Health care spending for the hospitalization, treatment, and complications of VTE is substantial, and rising faster than inflation [2]. Diagnosis and management of VTE remains a complicated endeavor because the majority are asymptomatic and evidence for routine screening is mixed [3]. Mechanical and chemical thromboprophylaxis have been proven to decrease the risk of VTE [4], and current guidelines recommend the use of chemical thromboprophylaxis, particularly enoxaparin, in trauma patients [5, 6]. However, significant practice variability exists based on the location of injury, estimated risk of bleeding, and varying institutional methods of risk stratification to guide therapy [3, 4, 7].

Several risk assessment models (RAMs) have been developed to stratify patients at risk of VTE. One validated model is the Caprini RAM, initially developed in 1991, and subsequently updated in 2013 and 2019 [8, 9]. Primarily created for the assessment of VTE risk in general surgical patients, the model has since been assessed for its applicability for specific inpatient populations including surgical critical care, as well as specialty surgical populations [10–14]. Despite its widespread use, there are several challenges associated with the practical application of the Caprini RAM. The current validated model contains 31 clinical variables, is time-consuming and includes costly laboratory tests [15]. Also, several variables necessitate a physical exam, which not only requires additional time to execute, but also allows for inter-rater variability [7]. These challenges are exacerbated in resource-limited settings, where standardized utilization of the model could be complicated by strained infrastructure and workforce demands. One study from a tertiary center in South Africa suggested that a low rate of thromboprophylaxis prescription may be due to a high burden of acute surgical and trauma patients, and time constraints for staff to compute risk scores [16].

Our aim is to determine if an abbreviated model could perform similarly to the Caprini RAM for reliable VTE risk assessment in trauma patients. We hypothesized that an abbreviated model would have similar discriminatory ability as the full Caprini RAM.

Methods

Study population

Trauma patients hospitalized in our American College of Surgeons-verified Level I Trauma Center between January 1, 2018, and December 31, 2018 were included if they met the following criteria: (1) age 18 years or older, (2) hospitalization longer than 24 h, and (3) no venous thromboembolism (VTE) identified at admission. We excluded patients who were made comfort measures only within 24 h of presentation. This study was reviewed and approved by the University of Massachusetts Medical School Institutional Review Board.

Primary outcome

The primary outcome was the development of inpatient VTE, defined as either a deep vein thrombosis (DVT) or pulmonary embolus (PE). Identification of VTE was based on clinical suspicion, with diagnostic testing ordered for confirmation. The presence of PE was confirmed by computed tomography angiography or ventilation-perfusion scan, and DVT by computed tomography angiography or ultrasonography. Our institution does not practice routine VTE surveillance given the lack of strong evidence to support this practice [3, 5, 6, 17, 18].

Covariates

The following demographic and clinical characteristics were analyzed: patient age in years, body mass index (BMI), sex, admission serum creatinine in mg/dL, Injury Severity Score (ISS), placement of epidural catheter, the mean number of days until VTE prophylaxis was initiated, initial VTE prophylaxis type (intermittent pneumatic compression [IPC] only, subcutaneous heparin, enoxaparin or other), if a patient was intubated, if there were any missed doses of VTE chemoprophylaxis at any point during the hospitalization, and blood product transfusion.

VTE prophylaxis protocol

The trauma surgery service at our institution employs a standardized protocol for VTE prophylaxis, with mechanical prophylaxis with IPC if injury does not preclude placement, and subcutaneous enoxaparin as the first line chemoprophylaxis agent. Dosing is weight-based, and subsequent dose adjustments are based upon anti-Xa levels [19]. Patients who are not eligible for enoxaparin due to renal insufficiency or presence of epidural catheters are started on subcutaneous heparin. Those who are ineligible for any chemical prophylaxis (e.g., intracranial hemorrhage, spinal cord injury, ongoing hemorrhage, or high risk of hemorrhage) receive mechanical prophylaxis with IPC, and are referred for placement of an inferior vena cava (IVC) filter if felt to be high risk for VTE or prolonged contraindication to chemoprophylaxis is anticipated.

Caprini score calculations

Caprini scores were calculated based upon the 2019 Caprini definitions [8]. There was no definition for the 5-point component of "serious trauma" within this update. Thus, we defined serious trauma as injuries either involving any two long bones (humerus, radius/ulna, femur, tibia/fibula), or involving any two of the Abbreviated Injury Scale (AIS) body regions. Injuries with an AIS score of less than two were excluded from consideration. This definition was used since it captures significant injuries by using an AIS score threshold, as well as injuries involving multiple body regions. A leg fracture for Caprini scoring was defined as any fracture involving the femur, tibia, fibula, or tarsals based on the Caprini 2019 update; fractures involving metatarsals or phalanges were not included [8].

Analytical approach

Analyses were performed in Stata 16.1 statistical software (StataCorp; College Station, Texas). The study population was first stratified by the presence or absence of VTE. Variables were described using Chi-Square and Student's t-tests for categorial and continuous variables, respectively. Fisher's exact test was used for categorical variables with any cell size of less than five. Caprini score components as well as mean Caprini scores were analyzed in a similar manner. All statistical tests were two-sided, with statistical significance defined as a p-value of <0.05. A single ideal cut point was determined by identifying the value that maximizes the product of the sensitivity and specificity [20]. Odds ratios (OR) with 95% confidence intervals (CI) were estimated using logistic regression models to examine the association of individual Caprini components and inpatient VTE. Any of the original Caprini RAM variables that met a predefined p-value threshold of <0.05 were used to calculate a new, abbreviated Caprini score. Logistic regression models were utilized to evaluate the odds of inpatient VTE and the original Caprini score, as well as the odds of VTE and the abbreviated Caprini score. Receiver operating characteristic (ROC) curves and c-statistics were calculated to assess each model's discriminatory ability.

Results

Study sample characteristics

The study sample was comprised of 1279 trauma patients. There were a total of 22 DVT and 14 PE in 33 patients, resulting in a confirmed VTE rate of 2.6%. There were three patients who developed both DVT and PE during their hospitalization. Compared to patients without VTE, those with VTE had higher mean BMI, Caprini score, ISS, and were more frequently intubated (Table 1). The two groups were otherwise similar regarding age, sex, days until initiation of VTE prophylaxis, initial VTE prophylaxis type, and number of missed doses, if any. The most common Caprini

Table 1 Characteristics of adults hospitalized for traumatic injury, stratified by the presence of VTE

	VTE (n = 33)		No VTE (n = 1246)		p-value
	Mean	SD	Mean	SD	
Age (years)	52.5	22.3	59.9	22.2	0.06
BMI	30.2	6.9	27.4	6.5	0.019
Days until initiation of VTE prophylaxis	0.90	1.0	0.88	1.0	0.89
Serum creatinine (mg/dL)	1.08	0.2	1.05	0.5	0.49
Caprini score	16.4	8.2	9.8	7.6	< 0.001
ISS	17.8	11.0	12.6	7.4	0.011
	n	%	*n*	%	
Epidural catheter placement	4	12.1	113	9.2	0.54
Initial VTE prophylaxis type					
Enoxaparin	9	27.3	548	44.5	0.14
IPC only	13	39.4	311	25.2	
Subcutaneous heparin	10	30.3	339	27.5	
Other	1	3.0	34	2.8	
Intubated	15	45.5	157	12.6	< 0.001
Male sex	22	66.7	761	61.1	0.52
Missed at least one dose of VTE chemoprophylaxis	14	42.4	344	28.3	0.08

BMI body mass index, *IPC* intermittent pneumatic compression, *ISS* Injury Severity Score, *SD* standard deviation, *VTE* venous thromboembolism score components within the VTE group were BMI ≥ 25 (69.7%), serious trauma (63.6%), and major surgery (60.6%) (Table 2).

Logistic regression analyses of the original and abbreviated Caprini RAMs

Unadjusted logistic regression identified which individual Caprini RAM components were most highly associated with the development of VTE while hospitalized (Table 3). From the original Caprini RAM, ten variables met the p-value threshold and were included in the abbreviated model: recent major surgery, length of surgery greater than 2 h, transfusion, restricted mobility greater than 72 h, central venous catheter, current major surgery, age, history of VTE, hip or leg fracture, and serious trauma. The median score for the original model was 8 (interquartile range [IQR] 5–13), and for the abbreviated model was 6 (IQR 3–10). The original and abbreviated Caprini RAMs were used in logistic regression models to evaluate the odds of inpatient VTE. The original RAM had an OR of 1.07 for each increase in Caprini score (95% CI 1.04–1.10), with a c-statistic of 0.753, sensitivity of 0.76, and specificity of 0.61. The abbreviated RAM had an OR of 1.17 for each increase in Caprini score (95% CI 1.10–1.25), with a c-statistic of 0.747, sensitivity of 0.73, and specificity of 0.62 (Table 4). Figure 1 illustrates the ROC curve for the original Caprini RAM, and Fig. 2 illustrates the abbreviated RAM. The optimal cut point for the original Caprini was 9, and for the abbreviated model was 7.

Discussion

Our analysis suggests that an abbreviated Caprini RAM has similar discriminatory ability to the original Caprini RAM in this sample of trauma patients. The abbreviated model was based on selection of only those items from the original Caprini RAM that were most highly associated with development of inpatient VTE. The two models shared similar median Caprini scores, odds ratios for VTE, c-statistics, sensitivity, and specificity, suggesting similar performance in ability to assess risk of VTE in trauma patients.

For a high volume, high acuity service with unpredictable demands such as trauma surgery, increasing workflow efficiency and decreasing administrative burden for clinicians is critical [21, 22]. As it stands, the original Caprini RAM has 31 individual items, several of which require a physical exam to assess. By reducing the model to ten items, all of which can be scored based on objective data, the time for completion is substantially reduced. Removing the majority of historical questions is also advantageous in the trauma population for two reasons: (1) many trauma patients are registered anonymously until identity is confirmed, making their medical charts initially inaccessible; and (2) critical status renders patients unable to

Table 2 Prevalence of Caprini score components, stratified by the presence of VTE

	VTE (n = 33)		No VTE (n = 1246)		p-value
Caprini 1-point section	n	%	n	%	
Active smoking	7	21.2	344	27.6	0.42
Acute myocardial infarction	0	0.0	19	1.5	1.00
Age 41–60 years	7	21.2	301	24.2	0.18
BMI > 25	23	69.7	753	60.4	0.28
BMI > 40*	1	3.0	60	4.8	1.00
Chemotherapy	1	0.3	12	0.1	0.29
Congestive heart failure	2	6.1	80	6.4	1.00
Diabetes mellitus requiring insulin	2	6.1	70	5.6	0.71
History of inflammatory bowel disease	0	0.0	17	1.4	1.00
History of prior major surgery	2	6.1	8	0.6	0.026
History of unexplained stillborn infant or recurrent spontaneous abortion	0	0.0	1	0.1	1.00
Length of surgery > 2 h**	17	51.5	274	21.9	< 0.001
Minor surgery	0	0.0	67	5.4	0.41
Oral contraceptives or hormone therapy	0	0.0	14	1.1	1.00
Pregnant or < 1 month postpartum	0	0.0	4	0.3	1.00
Restricted mobility < 72 h	12	36.4	337	27.1	0.24
Serious infection	3	9.1	40	3.2	0.09
Serious lung disease	3	9.1	121	9.7	1.00
Swollen legs	2	6.1	42	3.4	0.32
Transfusion	15	45.5	225	18.1	< 0.001
Varicose veins	0	0.0	3	0.2	1.00
Caprini 2-point section					
Age 61–74 years	7	21.2	257	20.6	0.18
Central venous catheter	15	50.0	133	10.9	< 0.001
History of malignancy	2	6.1	158	12.7	0.42
Immobilizing plaster cast	0	0.0	42	3.4	0.62
Major surgery	20	60.6	369	29.8	< 0.001
Restricted mobility > 72 h	18	54.6	316	25.4	< 0.001
Caprini 3-point section					
Age ≥ 75 years	6	18.2	386	31.0	0.18
Family history of VTE	0	0.0	10	0.8	1.00
History of VTE	6	18.2	72	5.8	0.003
Personal or family history of blood test that increases VTE risk	0	0.0	13	1.0	1.00
Caprini 5-point section					
Acute spinal cord injury with paralysis	1	3.0	14	1.1	0.33
Hip or knee arthroplasty	1	3.0	9	0.7	0.23
Hip, pelvis, or leg fracture	14	42.4	273	21.9	0.005
Serious trauma	21	63.6	552	44.3	0.027
Stroke	1	3.0	9	0.7	0.23

BMI body mass index, *VTE* venous thromboembolism

*BMI > 40 is additive with a BMI > 25 (i.e., 1 point for BMI > 25 and 1 point for BMI > 40)

**Length of surgery > 2 h is additive with major surgery

Table 3 Logistic regression models assessing inpatient VTE odds by individual Caprini components

	Odds ratio	p-value	95% Confidence interval	
Caprini 1-point section				
Active smoking	0.71	0.42	0.30	1.64
Acute myocardial infarction	No observations with acute myocardial infarction and VTE			
Age 41–60 years	0.54	0.20	0.21	1.37
BMI > 25	1.51	0.29	0.71	3.19
BMI > 40*	0.62	0.64	0.08	4.60
Chemotherapy	3.21	0.27	0.41	25.47
Congestive heart failure	0.94	0.93	0.22	4.00
Diabetes mellitus requiring insulin	1.08	0.91	0.25	4.62
History of inflammatory bowel disease	No observations with inflammatory bowel disease and VTE			
History of prior major surgery	9.98	0.005	2.04	48.96
History of unexplained stillborn infant or recurrent spontaneous abortion	No observations with recurrent abortion and VTE			
Length of surgery > 2 h**	3.77	< 0.001	1.88	7.56
Minor surgery	No observations with minor surgery and VTE			
Oral contraceptives or hormone therapy	No observations with oral contraceptive use and VTE			
Pregnant or < 1 month postpartum	No observations with postpartum and VTE			
Restricted mobility < 72 h	1.54	0.24	0.75	3.17
Serious infection	3.02	0.08	0.88	10.29
Serious lung disease	0.93	0.91	0.28	3.09
Swollen legs	1.85	0.41	0.43	7.98
Transfusion	3.78	< 0.001	1.88	7.62
Varicose veins	No observations with varicose veins and VTE			
Caprini 2-point section				
Age 61–74 years	0.63	0.34	0.25	1.61
Central venous catheter	2.86	< 0.001	1.98	4.14
History of malignancy	0.67	0.27	0.32	1.37
Immobilizing plaster cast	No observations with cast and VTE			
Major surgery	1.91	< 0.001	1.34	2.73
Restricted mobility > 72 h	1.88	< 0.001	1.33	2.66
Caprini 3-point section				
Age ≥ 75 years	0.36	0.041	0.14	0.96
Family history of VTE	No observations with family history of VTE and VTE			
History of VTE	1.54	0.006	1.13	2.08
Personal or family history of blood test that increases VTE risk	No observations with history of blood test that increases VTE risk and VTE			
Caprini 5-point section				
Acute spinal cord injury with paralysis	1.22	0.34	0.81	1.85
Hip or knee arthroplasty	1.34	0.17	0.88	2.04
Hip, pelvis, or leg fracture	1.21	0.007	1.05	1.40
Serious trauma	1.17	0.031	1.01	1.35
Stroke	1.34	0.17	0.88	2.04

BMI body mass index, *VTE* venous thromboembolism

*BMI > 40 is additive with a BMI > 25 (i.e., 1 point for BMI > 25 and 1 point for BMI > 40)

**Length of surgery > 2 h is additive with major surgery

Table 4 Logistic regression models assessing Caprini RAM and abbreviated Caprini RAM association with inpatient VTE odds

	Odds ratio	95% CI	C-statistic	Sensitivity	Specificity
Caprini RAM	1.07	1.04–1.10	0.753	0.76	0.61
Abbreviated Caprini RAM	1.17	1.10–1.25	0.747	0.73	0.62

CI Confidence interval, *RAM* risk assessment model, *VTE* venous thromboembolism

participate in an interview (due to intoxication, traumatic brain injury, injury severity, intubation, and/or sedation). As prior studies have reported significant scoring variability among clinicians in using VTE RAMs, the removal of subjective components in our abbreviated model may also increase inter-rater reliability [7, 23]. Lastly, the inclusion of only objective data may facilitate automated calculation of VTE risk by electronic health record (EHR) systems [11]. Physician-calculated scores have been shown to underestimate risk stratification when compared to computer-generated scores, leading to inadequate prophylaxis management and associated increased risk of VTE [24]. Therefore, a streamlined workflow and automated EHR clinical decision support may result in overall improved VTE prophylaxis prescribing practices [23, 25, 26].

The benefits of employing an abbreviated Caprini RAM may also extend to resource-limited settings. Institutions that operate with limited resources rely on over-burdened house staff and auxiliary healthcare workers [16]. Reducing time spent on cumbersome RAMs by using an abbreviated model, and also by facilitating automation in EHRs (if available), may increase the overall completion rate in these settings. In addition, the availability of specialized serum markers to assess the family history of hypercoagulable disorders for the original Caprini RAM (e.g., factor V Leiden, homocysteine, anti-cardiolipin antibodies, lupus anti-coagulant, prothrombin 20210A) [27], is highly variable and can be cost prohibitive [28]. The consistent use of uncommon laboratory markers for non-communicable disease is also subject to strained infrastructure and supply chain management. A lack of widely available testing then forces reliance on centralized laboratory processing, resulting in prolonged turnaround times if even accessible [29, 30], limiting its usefulness in time-sensitive circumstances. Our abbreviated Caprini RAM removes these costly laboratory tests, which may increase applicability in resource-limited settings.

The results of this study, although promising, require further assessment as this study represents a sample from a single tertiary care center. Although the Caprini RAM has been validated in a variety of patient populations [10, 12–14], there is limited data regarding its use in trauma patients [11]—a population with substantially high VTE risk [1, 31, 32]. Other RAMs for the trauma population have been studied, for instance the Trauma Embolic Scoring System, though the additional consideration of overall

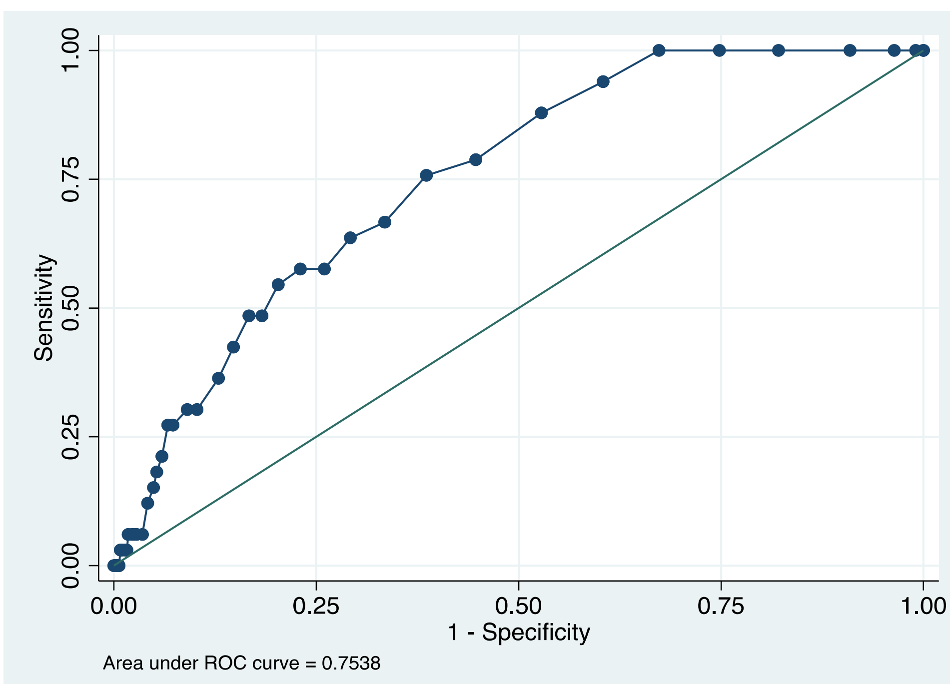

Fig. 1 Receiver operating characteristic curve of the Caprini RAM and inpatient VTE. *RAM* risk assessment model, *ROC* receiver operating characteristic, *VTE* venous thromboembolism

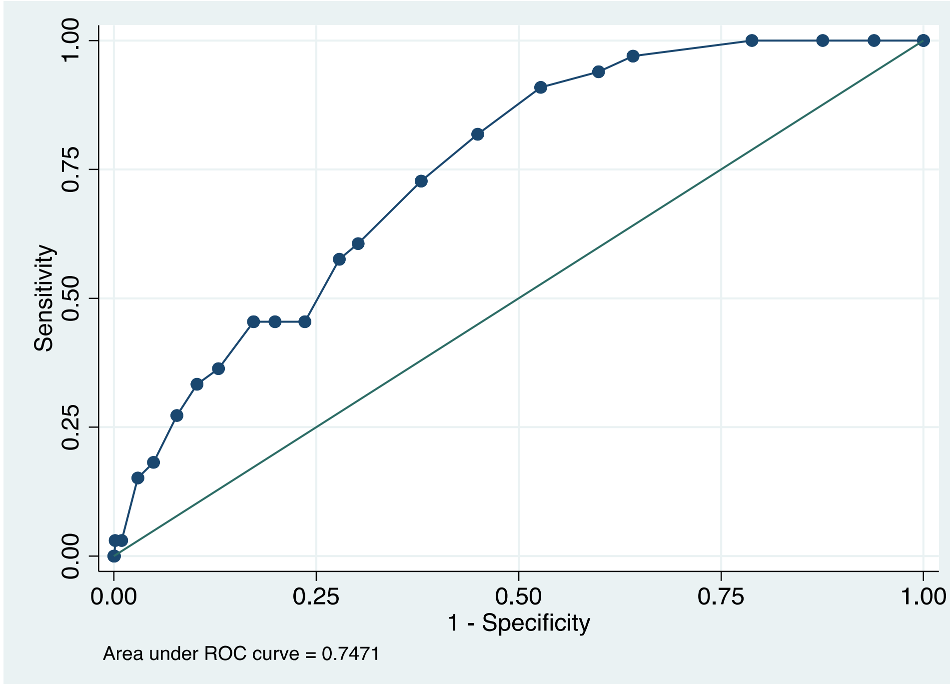

Fig. 2 Receiver operating characteristic curve of the abbreviated Caprini RAM and inpatient VTE. *RAM* risk assessment model, *ROC* receiver operating characteristic, *VTE* venous thromboembolism

ventilation days limits its utility in acute settings, when thromboprophylaxis is initiated within the first 24 h of presentation [33–35]. In contrast, the Caprini RAM is more practical for real time VTE risk assessment, as all required data is readily accessible by EHR review or patient interview and examination. Other considerations that are specific to trauma patients also include the management of injured patients at high risk for bleeding (and contraindicated for chemoprophylaxis). Thus, further study should seek to explore VTE risk assessment and decision-making algorithm of referral for placement of IVC filter.

With respect to diagnosing VTE, this study included only clinically suspected inpatient VTE, which likely underestimates the true incidence. Our institution does not use a standardized screening protocol for all patients, as evidence for routine surveillance is mixed. The Eastern Association for the Surgery of Trauma recommends surveillance only in high-risk patients [6], while the Western Trauma Association guidelines suggests consideration of ultrasound weekly depending on VTE risk and contraindications to chemoprophylaxis [5]. Furthermore, the American College of Chest Physicians recommends against routine screening in major trauma patients [3, 17, 18]. Additionally, this study included diagnosed VTE during index hospitalization only, and potentially excludes development of VTE past the date of discharge or in the case of readmission [36].

Conclusions

An abbreviated Caprini RAM has with similar discriminatory ability to the original Caprini RAM in trauma patients. The removal of subjective variables may increase workflow efficiency and reduce inter-rater variability. Additionally, the savings in terms of both cost and time may enhance utilization of such a model in settings with limited resources and workforce capacity. Future studies should consider multi-center investigations, and also validation in varying surgical populations to increase generalizability. An abbreviated model warrants further research as it has the potential to greatly impact workflow efficiency, which may lead to improved prescribing practices and overall decreased incidence of VTE.

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Data availability All data are available upon request.

Declarations

Conflict of interest No disclosures.

Ethical approval This study was reviewed and approved by the University of Massachusetts Medical School Institutional Review Board, with waiver of consent.

References

1. Roger FB (2001) Venous thromboembolism in trauma patients: a review. Surgery 130(1):1–12. https://doi.org/10.1067/msy.2001.114558
2. Fernandez MM, Hogue S, Preblick R, Kwong WJ (2015) Review of the cost of venous thromboembolism. Clinicoecon Outcomes Res 7:451–462. https://doi.org/10.2147/ceor.S85635
3. Rappold JF, Sheppard FR, Carmichael Ii SP et al (2021) Venous thromboembolism prophylaxis in the trauma intensive care unit: an American association for the surgery of trauma critical care committee clinical consensus document. Trauma Surg Acute Care Open 6(1):e000643. https://doi.org/10.1136/tsaco-2020-000643
4. Barrera LM, Perel P, Ker K, Cirocchi R, Farinella E, Morales Uribe CH (2013) Thromboprophylaxis for trauma patients. Cochrane Database Syst Rev. https://doi.org/10.1002/14651858.CD008303.pub2
5. Ley EJ, Brown CVR, Moore EE et al (2020) Updated guidelines to reduce venous thromboembolism in trauma patients: a western trauma association critical decisions algorithm. J Trauma Acute Care Surg 89(5):971–981. https://doi.org/10.1097/ta.0000000000002830
6. Rogers FB, Cipolle MD, Velmahos G, Rozycki G, Luchette FA (2002) Practice management guidelines for the prevention of venous thromboembolism in trauma patients: the EAST practice management guidelines work group. J Trauma 53(1):142–164. https://doi.org/10.1097/00005373-200207000-00032
7. Vyas D (2012) Variations in risk assessment models may contribute to the existing gap between venous thromboembolism prophylaxis guidelines and adherence. Springerplus 1(1):60. https://doi.org/10.1186/2193-1801-1-60
8. Cronin M, Dengler N, Krauss ES et al (2019) Completion of the updated caprini risk assessment model (2013 version). Clin Appl Thromb Hemost 25:1076029619838052. https://doi.org/10.1177/1076029619838052
9. Caprini JA, Arcelus JI, Hasty JH, Tamhane AC, Fabrega F (1991) Clinical assessment of venous thromboembolic risk in surgical patients. Semin Thromb Hemost 17(Suppl 3):304–312
10. Obi AT, Pannucci CJ, Nackashi A et al (2015) Validation of the Caprini venous thromboembolism risk assessment model in critically ill surgical patients. JAMA Surg 150(10):941–948. https://doi.org/10.1001/jamasurg.2015.1841
11. He L, Luo L, Hou X et al (2021) Predicting venous thromboembolism in hospitalized trauma patients: a combination of the Caprini score and data-driven machine learning model. BMC Emerg Med 21(1):60. https://doi.org/10.1186/s12873-021-00447-x
12. Krauss ES, Segal A, Cronin M et al (2019) Implementation and validation of the 2013 Caprini score for risk stratification of arthroplasty patients in the prevention of venous thrombosis. Clin Appl Thromb Hemost 25:1076029619838066. https://doi.org/10.1177/1076029619838066
13. Pannucci CJ, Bailey SH, Dreszer G et al (2011) Validation of the Caprini risk assessment model in plastic and reconstructive surgery patients. J Am Coll Surg 212(1):105–112. https://doi.org/10.1016/j.jamcollsurg.2010.08.018
14. Shuman AG, Hu HM, Pannucci CJ, Jackson CR, Bradford CR, Bahl V (2012) Stratifying the risk of venous thromboembolism in otolaryngology. Otolaryngol Head Neck Surg 146(5):719–724. https://doi.org/10.1177/0194599811434383
15. Mlaver E, Lynde GC, Gallion C, Sweeney JF, Sharma J (2020) Development of a novel preoperative venous thromboembolism risk assessment model. Am Surg 86(9):1098–1105. https://doi.org/10.1177/0003134820943556
16. Rocher WD, Page T, Rocher M, Nel D (2019) Venous thromboembolism risk and prophylaxis prescription in surgical patients at a tertiary hospital in Eastern Cape Province, South Africa. S Afr Med J 109(3):178–181. https://doi.org/10.7196/SAMJ.2019.v109i3.13510
17. Gould MK, Garcia DA, Wren SM et al (2012) Prevention of VTE in nonorthopedic surgical patients: antithrombotic therapy and prevention of thrombosis, 9th ed: American college of chest physicians evidence-based clinical practice guidelines. Chest 141(2 Suppl):e227S – e277. https://doi.org/10.1378/chest.11-2297
18. Kodadek LMHE (2016) Screening and diagnosis of VTE: the more you look, the more you find? Curr Trauma Rep 2(2):29–34
19. Bigos R, Solomon E, Dorfman JD, Ha M (2021) A weight- and anti-Xa-guided enoxaparin dosing protocol for venous thromboembolism prophylaxis in intensive care unit trauma patients. J Surg Res 265:122–130. https://doi.org/10.1016/j.jss.2021.02.034
20. Liu X (2012) Classification accuracy and cut point selection. Stat Med 31(23):2676–2686. https://doi.org/10.1002/sim.4509
21. Ramaniuk A, Dickson BJ, Mahoney S, O'Mara MS (2017) Maximizing efficiency on trauma surgeon rounds. J Surg Res 207:198–204. https://doi.org/10.1016/j.jss.2016.08.064
22. Sammann A, Chehab LZ, Patel D, Liao J, Callcut R, Knudson MM (2020) Improving efficiency and meeting expectations without compromising care on trauma surgical rounds. J Surg Res 247:163–171. https://doi.org/10.1016/j.jss.2019.10.026
23. Beck MJ, Haidet P, Todoric K, Lehman E, Sciamanna C (2011) Reliability of a point-based VTE risk assessment tool in the hands of medical residents. J Hosp Med 6(4):195–201. https://doi.org/10.1002/jhm.860
24. Pannucci CJ, Obi A, Alvarez R et al (2014) Inadequate venous thromboembolism risk stratification predicts venous thromboembolic events in surgical intensive care unit patients. J Am Coll Surg 218(5):898–904. https://doi.org/10.1016/j.jamcollsurg.2014.01.046
25. Amland RC, Dean BB, Yu H et al (2015) Computerized clinical decision support to prevent venous thromboembolism among hospitalized patients: proximal outcomes from a multiyear quality improvement project. J Healthc Qual 37(4):221–231. https://doi.org/10.1111/jhq.12069
26. Tooher R, Middleton P, Pham C et al (2005) A systematic review of strategies to improve prophylaxis for venous thromboembolism in hospitals. Ann Surg 241(3):397–415. https://doi.org/10.1097/01.sla.0000154120.96169.99
27. Caprini JA, Glase CJ, Anderson CB, Hathaway K (2004) Laboratory markers in the diagnosis of venous thromboembolism. Circulation 109(12 Suppl 1):I4-18. https://doi.org/10.1161/01.Cir.0000122869.59485.36
28. Bilgi K, Muthusamy A, Subair M et al (2016) Assessing the risk for development of venous thromboembolism (VTE) in surgical patients using adapted Caprini scoring system. Int J Surg 30:68–73. https://doi.org/10.1016/j.ijsu.2016.04.030
29. Kuupiel D, Bawontuo V, Drain PK, Gwala N, Mashamba-Thompson TP (2019) Supply chain management and accessibility to point-of-care testing in resource-limited settings: a systematic scoping review. BMC Health Serv Res 19(1):519. https://doi.org/10.1186/s12913-019-4351-3
30. Drain PK, Hyle EP, Noubary F et al (2014) Diagnostic point-of-care tests in resource-limited settings. Lancet Infect Dis 14(3):239–249. https://doi.org/10.1016/s1473-3099(13)70250-0
31. Bahloul M, Dlela M, Bouchaala K et al (2020) Post-traumatic pulmonary embolism: incidence, physiopathology, risk factors of early occurrence, and impact outcome. A narrative review. Am J Cardiovasc Dis 10(4):432–443
32. Geerts WH, Code KI, Jay RM, Chen E, Szalai JP (1994) A prospective study of venous thromboembolism after major trauma. N Engl J Med 331(24):1601–1606. https://doi.org/10.1056/nejm199412153312401

33. Greenfield LJ, Proctor MC, Rodriguez JL, Luchette FA, Cipolle MD, Cho J (1997) Posttrauma thromboembolism prophylaxis. J Trauma 42(1):100–103. https://doi.org/10.1097/00005373-199701000-00017
34. Rogers FB, Shackford SR, Horst MA et al (2012) Determining venous thromboembolic risk assessment for patients with trauma: the trauma embolic scoring system. J Trauma Acute Care Surg 73(2):511–515. https://doi.org/10.1097/ta.0b013e3182588b54
35. Ho KM, Rao S, Rittenhouse KJ, Rogers FB (2014) Use of the trauma embolic scoring system (TESS) to predict symptomatic deep vein thrombosis and fatal and non-fatal pulmonary embolism in severely injured patients. Anaesth Intensive Care 42(6):709–714. https://doi.org/10.1177/0310057x1404200605
36. Yale SH, Medlin SC, Liang H, Peters T, Glurich I, Mazza JJ (2005) Risk assessment model for venothromboembolism in post-hospitalized patients. Int Angiol 24(3):250–254